Design Proposal: Mach-Zehnder Interferometer

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Index Terms — Mach-Zehnder interferometer, silicon photonics.

I. Introduction

The Mach-Zehnder interferometer (MZI) serves as a fundamental tool for measuring relative phase shifts between two coherent light beams. Utilizing the MZI architecture, various optoelectronic modulators, including electro-optic and thermo-optic variants, have been developed. This paper introduces a specific MZI design and evaluates its performance through detailed simulation results.

II. Theory

Let us consider a symmetrical Y-junction splitter. An incident beam with an input intensity I_i and corresponding electric field E_i is divided equally between two arms. Consequently, the intensity and electric field in each branch are defined as $I_{1,2} = I_i/2$ and $E_{1,2} = E_i/\sqrt{2}$, respectively. Regarding the optical combiner, a single-branch input is distributed equally between the fundamental waveguide mode and higher-order radiation modes. Therefore, the resulting output field from a single arm is $E_{o,n} = E_n/\sqrt{2}$. When both arms are considered, the total output field E_o results from the vector superposition of the input fields, scaled by a factor of $1/\sqrt{2}$, yielding $E_o = (E_1 + E_2)/\sqrt{2}$. For a monochromatic plane wave, the electric field is represented by $E = E_0 e^{i(\omega t - \beta z)}$, where $\beta = 2\pi n/\lambda$ is the propagation constant and n represents the refractive index of the medium.

Assuming a lossless regime, the fields in the two arms can be expressed as

$$E_{o1} = E_1 e^{-i\beta L_1} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_1} \tag{1}$$

$$E_{o2} = E_2 e^{-i\beta L_2} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_2} \tag{2}$$

By substituting these into the expression for the total output field, we obtain:

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{2}\left(e^{-i\beta L_1} + e^{-i\beta L_2}\right) \tag{3}$$

The resulting output intensity, $I_o \propto |E_o|^2$, can then be derived as follows:

$$I_o = \frac{I_i}{4}\left|e^{-i\beta L_1} + e^{-i\beta L_2}\right|^2 = \frac{I_i}{2}[1 + \cos(\beta \Delta L)] \tag{4}$$

III. Modelling and simulation

A. Waveguide

The proposed design utilizes a strip waveguide with a cross-section of 500 nm width and 220 nm height. All devices are designed for TE mode propagation, with the polarization in the plane of the chip. The electric field intensity profiles for the fundamental quasi-TE$_0$ and quasi-TM$_0$ modes are simulated using Lumerical MODE Solutions and presented in Figures 1 and 2, respectively.

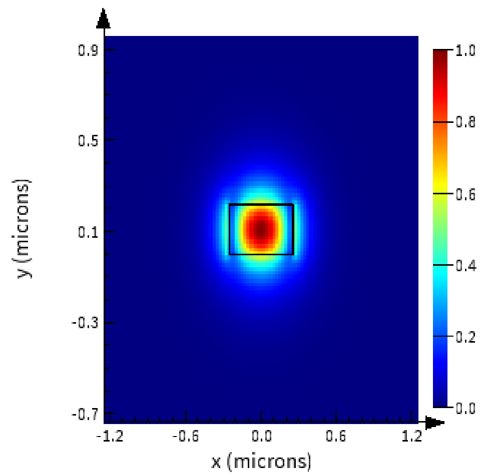

Fig. 1. E$_x$ component of electric field intensity for TE$_0$ mode for 1550nm.

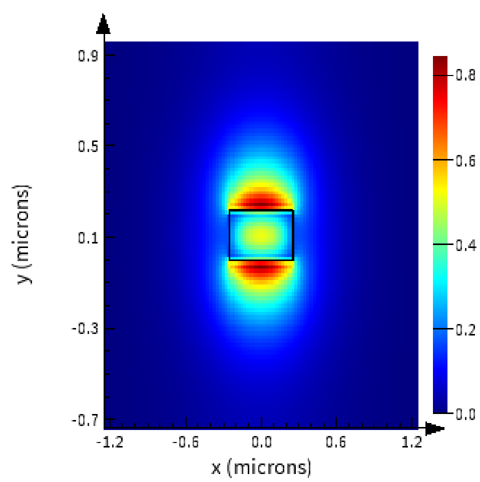

Fig. 2. E$_y$ component of electric field intensity for TM$_0$ mode for 1550nm.

Furthermore, the spectral dependence of the effective index (n_{eff}) and the group index (n_g) for TE mode are plotted in Figures 3 and 4, respectively. As observed, the effective index decreases as the wavelength increases, whereas the group index exhibits an upward trend over the same spectral range.

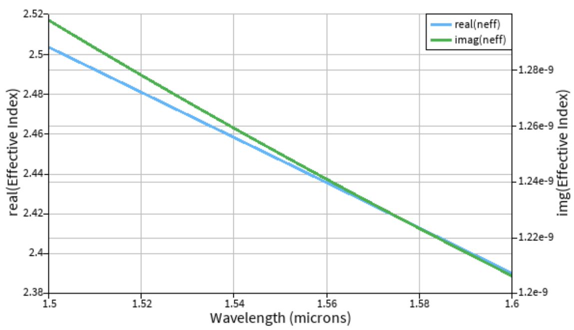

Fig. 3. Effective index of the waveguide versus wavelength.

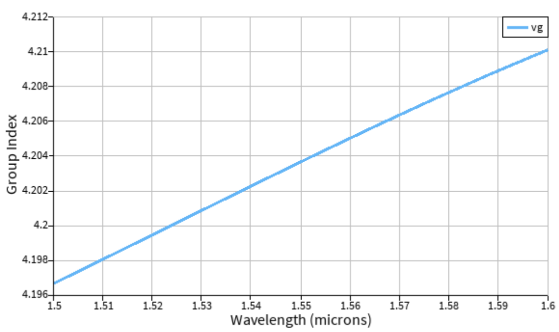

Fig. 4. Group index of the waveguide versus wavelength.

Using these data for n_{eff} and MATLAB fitting, a compact model was derived to characterize the waveguide's dispersion, yielding the following second-order polynomial fit:

$$n_{eff}(\lambda) = 2.4468 - 1.1333\,(\lambda - \lambda_0) - 0.0439\,(\lambda - \lambda_0)^2 \quad (5)$$

The corresponding graph is shown in Figure 5.

B. Mach-Zehnder Interferometer

For a lossless Mach-Zehnder interferometer, the transfer function as a function of wavelength (λ) is expressed as :

$$T_{MZI}(\lambda) = \frac{I_{out}}{I_{in}} = \frac{1}{2}[1 + \cos(\beta \Delta L)] \quad (6)$$

where $\Delta L = L_1 - L_2$ is the path length difference and $\beta = \frac{2\pi}{\lambda} n_{eff}(\lambda)$ is the propagation constant, assuming $\beta_1 = \beta_2$.
The spectral distance between the peaks of the MZI transfer function, known as the free spectral range (FSR), is given by:

$$FSR = \lambda^2 / (n_g \Delta L) \quad (7)$$

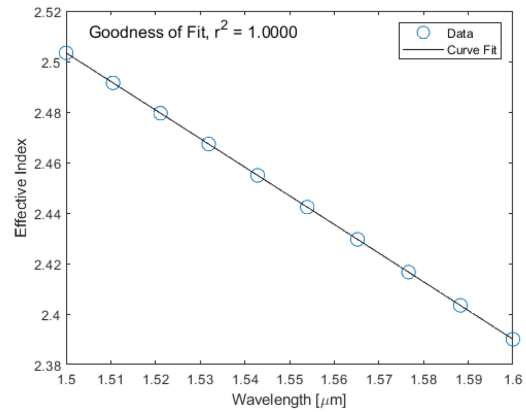

Fig. 5. Compact model for the waveguide mode TE_0.

where the group index, n_g, accounts for the dispersion of the waveguide and is defined as:

$$n_g = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda} \quad (8)$$

Table 1 lists the various path length differences ΔL for the designed MZI devices alongside the expected FSR values for each device, calculated for TE_0 mode at $\lambda = 1550nm$ and $n_g = 4.2$.

TABLE I
FSR with different ΔL.

$\Delta L(\mu$ m)	$FSR($ nm)
25	22.88
50	11.44
100	5.72
200	2.86
400	1.43
600	0.95

Next, we simulate the transmission spectrum of the MZI using Lumerical INTERCONNECT. We construct an interferometer circuit using Y-branch S-parameter elements from the course device library [Ref], and we add the TE-mode waveguide elements simulated by Lumerical MODE in the previous section. Figure 6 shows the simulation results for

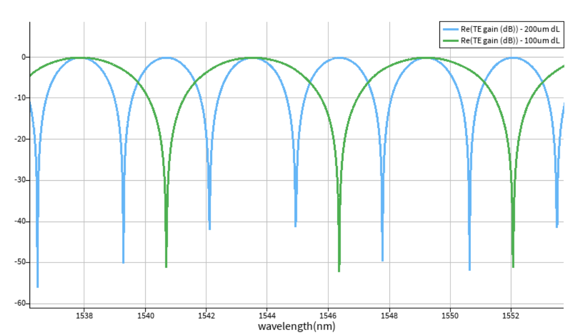

Fig. 6. Spectrum of MZI for two ΔL of $200\mu m$ and $100\mu m$.

two devices with path length differences ΔL of $200\mu m$ and $100\mu m$.

Then, we simulate the transmission spectrum by taking into account the insertion loss from two grating couplers, using the data provided from the course device library [Ref]. The resulting spectra are presented in Figure 7.

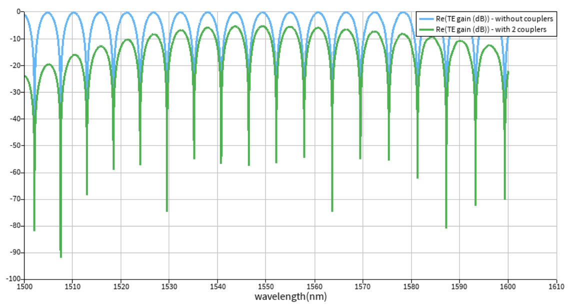

Fig. 7. Spectrum of MZI of $\Delta L = 100\mu m$ with and without couplers.

Figure 8 shows insertion loss from two grading couplers calculated using Lumerical INTERCONNECT.

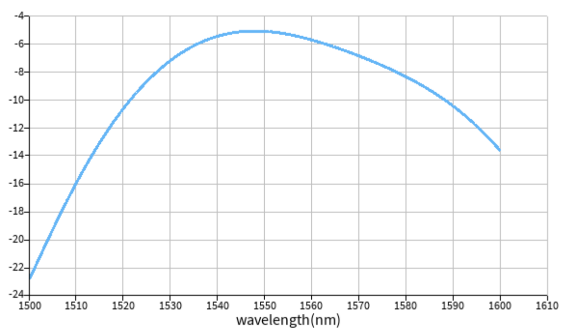

Fig. 8. Insertion loss of two grating couplers.

IV. Fabrication

K-Layout design. To be continue.

VI. Experimental Data

To be continue.

VII. Analysis

To be continue.

VIII. Conclusion

In this article, we propose several Mach-Zehnder interferometers with different ΔL using silicon strip waveguide. To be continue.